# State as Computation: Moving Structural Search between Representation Construction and Downstream Reasoning

ORION-P11

21 August 2026

**Abstract**

Reasoning systems are commonly compared as if the state presented to the downstream learner were a fixed observation. We study a different resource boundary: **constructing task-relevant state is itself computation**, and this computation can substitute for structural search performed downstream. For a query family `F`, any fixed representation supporting exact linear readout of every query requires dimension at least `rank(F)`. For all size-`s` parity queries on `d` Boolean variables this becomes `binom(d,s)` accessible coordinates, whereas a query-conditioned construction needs only the coordinates selected by the current query. Frozen controlled studies produce universal/compiled representation ratios of 91×–1820× and dense-decoder sample-threshold reductions from 4× to more than 32×. A no-answer-laundering construction exposes 5–7 latent components rather than the final label and retains large gains. A preregistered sparse universal decoder **falsifies** the stronger claim that compilation retains at least a 4× threshold advantage in both hostile cells, leaving 2× and 4× residual gaps. A fresh deterministic replication reproduces those residuals; a separately frozen deterministic single-thread ExtraTrees successor fails to reach 0.95 through `n=1024` where compiled state reaches it at `n=64`. Exact workload laws quantify option debt when specialized state is retained without raw recoverability. The result is not universal dominance of compilation: **state construction determines where structural-search computation is paid**, and the placement can be measured jointly through accessible rank, downstream burden, construction cost and future optionality.

## 1 Introduction

A system can possess all information required for a task and still make that information expensive for a bounded learner or reasoner to use. Computationally usable information formalizes one version of this observation. Partial evaluation and knowledge compilation move work upstream. Materialized views trade preprocessing and storage for later query cost. Current agent systems retrieve, compress, summarize or restructure context before generation. Recent LLM evidence also shows that state design itself can materially change dynamic reasoning while model parameters remain fixed.

Those results make several weak novelty claims untenable. P11 does **not** claim that representation matters, that computation can create usable information, that query-conditioned memory is new, or that compression can reduce downstream cost. The unresolved question is:

> **When task-relevant structure can be discovered either while constructing state or later by a decoder/search process, where is the computation paid, how much downstream burden can be removed, and what future optionality is lost by specialization?**

We call this view *state as computation*. A compiler `C(R,q)` receives raw state `R` and query `q`, constructs task-facing state, and hands it to a bounded downstream access mechanism. Compilation is never free: compiler operations, state bytes, training cost, cache/recovery cost, downstream samples/search, verifier/tool calls, latency and reproducible energy belong to one resource receipt.

The paper contributes accessible-rank theory, controlled compilation gaps, hostile decoder substitution and future-optionality laws. The strongest conclusion is mechanistic: **representation construction and downstream access are two loci at which structural-search work can be paid**.

# 2 Donor boundary and novelty residual

Predictive `V`-information already establishes that computational constraints change what information is usable and that computation can transform unusable information into usable information. Classical partial evaluation specializes programs to known inputs. Knowledge compilation and database materialization move computation upstream for later reuse. Feature selection and sparse models search for relevant coordinates inside a larger representation. Query-conditioned memory and retrieval condition state on the current task. Long-horizon context-compression systems explicitly optimize memory/performance trade-offs. Wong et al. provide direct evidence that state representation and construction can change LLM reasoning.

P11 subtracts all of those primitives.

The residual is a **joint placement account**:

`raw state -> construction work -> task-facing state -> decoder/search work -> verified outcome`

with a future horizon:

`task-facing state + raw/cache policy -> future query service or option debt.`

The paper asks how accessible rank, downstream sample/search burden, upstream construction, cache/recovery and future-query coverage move together when the same underlying information is exposed differently.

# 3 Accessible-rank theory

Let `X` be a domain with distribution `mu`, and let `F={f_1,...,f_N}` be query functions in `L2(mu)`. A fixed query-agnostic representation is `phi:X->R^m`. It supports exact linear query answering when, for every `q`, some `w_q` satisfies `f_q(x) = <w_q, phi(x)>` almost surely.

## 3.1 Query-family accessible-rank bound

If `span(F)` has dimension `r`, every fixed representation supporting exact linear readout of all queries has `m >= r`.

**Proof.** Every `f_q` lies in the span of the `m` coordinate functions of `phi`. That span has dimension at most `m` and must contain the `r`-dimensional span of `F`; hence `m>=r`. The theorem is elementary linear algebra; its role here is as a systems resource boundary, not as a claim of mathematical novelty.

## 3.2 Approximate orthonormal frontier

For orthonormal `f_1,...,f_N` and any `m`-dimensional linearly accessible subspace `U`, Bessel's inequality gives `(1/N) sum_q ||f_q - P_U f_q||_2^2 >= 1 - m/N`. This is an access-class statement, not a lower bound on unrestricted nonlinear decoding.

### 3.3 Parity corollary

For `X={-1,+1}^d` under the uniform measure and all size-`s` subsets `S`, define `f_S(x)=product_{i in S} x_i`. Distinct parity characters are orthogonal. A fixed exact linear-accessible representation supporting all size-`s` queries therefore requires at least `binom(d,s)` coordinates. A query-conditioned construction need expose only the selected query structure. This establishes an exact accessible-representation gap, not a total-time lower bound.

## 4 Dense controlled studies

The confirmatory experiment compares raw linear input, a fixed universal parity bank, and query-conditioned compiled state over a frozen train-size grid.

| `d` | `s` | universal dims | compiled dims | universal/compiled | compiled `n` at 0.90 | universal `n` at 0.90 |
|---|---|---|---|---|---|---|
| 14 | 2 | 91 | 1 | 91× | 32 | 128 |
| 16 | 4 | 1820 | 1 | 1820× | 32 | `NOT_REACHED` by 1024 |
| 18 | 3 | 816 | 1 | 816× | 32 | 1024 |
| 20 | 3 | 1140 | 1 | 1140× | 32 | 1024 |

At `n=1024`, compiled accuracy is 1.0 in every cell. Raw linear remains near chance. The result is consistent with a structural-search interpretation: decisive coordinates exist in the universal bank, but a bounded dense decoder must identify them among many irrelevant candidates.

### 4.1 No-answer-laundering control

P11B exposes only the `r` active parity components selected by a query, with `r` in {`5,7`}. The downstream logistic learner must still infer an odd-cardinality majority target; no compiled component equals or negates the final label.

| `d` | `s` | `r` | universal dims | universal `n` at 0.95 | compiled `n` at 0.95 |
|---|---|---|---|---|---|
| 15 | 3 | 5 | 455 | 2048 | 64 |
| 17 | 3 | 5 | 680 | 2048 | 64 |
| 17 | 4 | 5 | 2380 | `NOT_REACHED` by 2048 | 64 |
| 19 | 3 | 7 | 969 | `NOT_REACHED` by 2048 | 64 |

The high-dimensional cells therefore exhibit at least 32× threshold separation under the registered dense decoder without answer laundering.

## 5 Hostile decoder substitution

The central alternative explanation is that the universal representation is penalized only because the downstream decoder has the wrong inductive bias. If so, stronger decoder-side search should buy back the compilation advantage. P11 treats that prediction as a mechanism test.

### 5.1 P11D sparse decoder — permanent negative

P11D preregistered a strong hostile gate: an L1 sparse universal decoder should still leave at least a 4× sample-threshold advantage for compiled state in both high-dimensional cells. It did not.

| cell (d,s,r) | sparse universal n at 0.95 | compiled n at 0.95 | ratio | compiled - sparse at n=64 |
|---|---|---|---|---|
| (17,4,5) | 128 | 64 | 2× | +0.2903 |
| (19,3,7) | 256 | 64 | 4× | +0.3840 |

The terminal `P11D_SPARSE_DECODER_GAP_NOT_MET` remains permanently negative. P11D also exposed an unseeded `liblinear` replay defect; that defect is retained.

### 5.2 P11E deterministic sparse replication

P11E uses a fresh data seed and explicit estimator seeds. It reproduces thresholds 128/64 and 256/64 with low-sample compiled-minus-sparse advantages +0.2912 and +0.3307. Two fresh executions produce canonical SHA-256 `1097d94bef1132d4dfa5d01176a9fcfcfebc46de8113e7cb2e57da1e579a4536`.

### 5.3 P11C/P11F/P11G nonlinear sequence

P11C's first execution attempt exceeded the available window; after an amendment that vectorized only its parity-bank evaluation, `P11C_EXECUTION_RECEIPT_V1.md` records the frozen protocol run to completion twice at `P11C_STRONGER_DECODER_GAP_SUPPORTED`, with its pooled ≥4× gate passing at exactly the boundary and a twenty-seed sweep putting that boundary at 11 of 20 draws. It settles nothing and carries no claim authority; what it does establish is that its pooled combination rule was applied inside its own protocol. P11F produced positive numerical separation but is non-authoritative because hostile review found `n_jobs=-1` violated the frozen otherwise-default tree contract. P11G was frozen separately with `n_jobs=1`, explicit random states, and two fresh subprocess replays in the terminal decision path.

| cell | deterministic tree universal n at 0.95 | compiled n at 0.95 | tree accuracy at n=1024 | compiled - |
|---|---|---|---|---|
| (17,4,5) | NOT_REACHED | 64 | 0.8248 | |
| (19,3,7) | NOT_REACHED | 64 | 0.7828 | |

P11G's terminal is `P11G_DETERMINISTIC_TREE_DECODER_GAP_SUPPORTED`; both fresh scientific payloads have SHA-256 `a2b0c33ce3c39e54ca1aa400a2b7d52d019fc4503f6cd5eb726c7b8bbe79a7cc`.

### 5.4 What P11G's terminal is a statement about

The programme registered three universal-state arms in P11C — `UNIVERSAL_L2`, `UNIVERSAL_L1` and `UNIVERSAL_EXTRA_TREES` — and P11G's receipt publishes curves for one. Replaying P11G's own frozen data stream with only the decoder swapped, and reading P11G's own four scientific gates on each arm:

| universal arm | 0.95 threshold per cell, censored at 256 | terminal P11G's own gates p |
|---|---|---|
| `UNIVERSAL_L2` | ≥256, ≥256 | `..._GAP_SUPPORTED` |
| `UNIVERSAL_L1` | **128**, ≥256 | `..._GAP_NOT_MET` |
| `UNIVERSAL_EXTRA_TREES` (reported) | ≥256, ≥256 | `..._GAP_SUPPORTED` |

Two of three comparable pairs change the verdict, so the arm axis is not inert, and the flip is entirely the threshold gate: 128 is not ≥256. This is the same sparse threshold P11D reports as a permanent negative and P11E replicates; what is new is that placing it in P11G's gate, on P11G's own bytes, prints `NOT_MET`. `NOT_REACHED` through `n=1024` is therefore not a stronger reading than 128 — an arm that reaches nothing anywhere gives the same gate reading in every world.

P11G's terminal is retained exactly as frozen and is evidence about the decoder its own claim-authority sentence names. `P11G_ARM_PLACEMENT_ADJUDICATION_V1.md` carries the adjudication, including the finding — read off the two freezes — that P11C's pooled combination rule governs P11C and does not bind P11G.

### 5.5 Decomposing a gap that moves two things at once

P11G compares L2 logistic regression on `r` compiled columns against a 96-tree ensemble on the complete bank. Holding the decoder at ExtraTrees and moving only the representation separates them.

| cell | published gap at `n=64` | decoder-family half | state half | state share |
|---|---|---|---|---|
| (17,4,5) | +0.4624 | +0.0614 | +0.4010 | 86.7% |
| (19,3,7) | +0.3942 | +0.1757 | +0.2185 | 55.4% |

It cuts both ways and is reported both ways. It narrows the terminal: 13.3% and 44.6% of the published gaps are the change of decoder family, not of state. It supports the placement claim: with the decoder held fixed the state half is the majority in both cells, and being measured at a fixed decoder it is unaffected by which universal arm sits in the gate.

The sequence supports the interpretation that **compilation and decoder inductive bias are alternative locations for structural search**. Stronger downstream structure discovery should shrink the upstream advantage; the sparse negative is therefore part of the mechanism evidence, not an inconvenient result to erase. Each verdict in it is scoped to the arm that produced it.

### 5.6 P11H pooled successor — the survival was not predetermined, and it did not hold

The arm-scoping above is the smaller finding. The larger one, from `orion.study.p11.attack_audit`, is that all four of P11G's scientific gates hold in **every world its own freeze admits**, so its survival was fixed before its seed was drawn. P11H re-asks the question under a protocol whose attack can win, editing nothing of P11G.

P11H registers all three universal arms and freezes the best-of-arms rule inside its own positive gate; carries P11G's `0.95` and `0.20` thresholds over unedited; and draws two protected regimes by its fresh seed from a frozen 2×3 ladder of state widths `r` ∈ {`3, 7`} crossed with complete parity banks of 91, 364 and 969 columns. Before execution, the recorded preflight reports both hypothesis gates `BOTH_OUTCOMES_REACHABLE` — supports `[0.8808, 1.0000]` against `0.95` and `[0.0000, 0.2482]` against `0.20` — and two reachable terminals over 15 admissible draws, 3 of which clear every gate.

The seed drew `(14,2,3)` and `(19,3,3)`. Every precondition held, replay was byte-identical, and both hypothesis gates failed: `P11H_POOLED_UNIVERSAL_ATTACK_PREVAILED`.

| rung | universal bank | pooled 0.95 threshold | `delta64` |
|---|---|---|---|
| (14,2,3) | 91 | **128** | +0.1482 |
| (14,3,3) | 364 | **128** | +0.0992 |
| (19,3,3) | 969 | **128** | +0.0506 |
| (14,2,7) | 91 | ≥256 | +0.2350 |
| (14,3,7) | 364 | ≥256 | +0.3172 |
| (19,3,7) | 969 | ≥256 | +0.3175 |

The pool reaches the target by `n=128` at every `r=3` rung and at no `r=7` rung, while the complete bank moves 91 → 969 columns inside each half without changing a verdict. The advantage is governed by the width of the compiled state, not by the size of the universal representation.

And the decomposition sharpens in the opposite direction: at the drawn regimes the decoder-family half of the `n=64` gap is exactly `+0.0000`, so **100%** of it is the change of state — against 86.7%

and 55.4% above — and the gap is still only +0.0506. Attribution and magnitude are different questions.

Three of the fifteen admissible draws would have printed the positive terminal and the seed did not draw one, so the r=7 rungs carry no terminal and no claim authority. The r=5 boundary was excluded before execution for verdict instability across three preflight seeds, in both directions, and remains open.

### 5.7 P11I wide high-width replication

P11I prospectively freezes the narrower regime P11H located rather than relabeling P11H. It evaluates the complete cross of three fresh execution seeds and three fixed bank-geometry strata at r=7, with a matched r=3 control in every cell. The independent random unit is the execution seed (n=3); geometry is a fixed within-seed stratum and five query repeats are technical repeats within each cell. One failed cell still defeats the conjunction.

All nine high-width units pass. Compiled accuracy at n=64 ranges 0.9690–0.9981; the pooled attack's best accuracy below n=256 ranges 0.8489–0.9421 against the strict <0.95 gate; and delta64 ranges +0.2463–+0.3543 against >=+0.20. The same pooled attack reaches 1.0000 in all nine matched r=3 controls. Two fresh subprocess payloads are byte-identical at SHA-256 b50ace30...e0ce.

The P11I terminal is P11I_HIGH_WIDTH_ADVANTAGE_REPLICATED_WIDE_PANEL. It licenses a width-conditioned result only: the pooled attack wins in the narrow regime and the compiled-state advantage replicates in the registered high-width regime. A fresh two-subprocess revalidation under P11I_REPLICATION_UNIT_AMENDMENT_V1_1.md reproduces every cell byte-identically while recording n=3, not nine. P11D and P11H remain adverse historical results.

## 6 Future optionality and accessibility-work accounting

Current accessibility and future optionality are different objectives. Let a universal state contain N independent query coordinates and a compiled state retain r.

If raw source is lost, one-step coverage for a uniformly random future query is exactly r/N. If K independently selected size-r compilations are cached, expected coverage is 1-(1-r/N)^K. Under uniform demand, the expected number of distinct requested components after K queries is N[1-(1-1/N)^K].

These laws produce four policy regimes: compile only; compile + cache; retain raw + compile; and universal materialization. In the frozen workload study, the first grid point where universal bulk compilation becomes cheaper than expected cache compilation occurs at 0.5N, 1.0N, and 2.0N future-query horizons for batch-efficiency factors 0.25, 0.50, and 0.75. Concentrated Zipf-like workloads delay crossover because fewer distinct coordinates are demanded.

Any comparison that treats representation construction as free is invalid. P11 therefore keeps separate receipt coordinates for compiler/preprocessing operations and latency, state bytes/tokens and memory traffic, downstream model identity/capacity, training examples or generated/recurrent steps, search nodes/verifier/tool calls, cache and reuse, raw recovery/reconstruction/recompilation, end-to-end latency and reproducible energy where available.

Learned-compiler training cost must be reported separately and amortized only under a prospectively stated reuse horizon. Unless an application supplies exchange rates, comparisons should be Pareto surfaces rather than post-hoc scalar costs.

## 7 Methods, reproducibility and related work

The controlled experiments use prospectively frozen seeds, cell grids, train-size grids, thresholds and hostile checks. Thresholds are observed first grid points and are never extrapolated beyond the registered maximum; `NOT_REACHED` is retained literally. Exact theorems and workload equations require no statistical inference.

The evidence history is append-only in scientific meaning: P11/P11B establish dense and no-answer-laundering gaps; P11D fails its stronger sparse gate; P11E independently reproduces the surviving residual; P11C, after a no-outcome vectorization amendment, ran to completion twice at `P11C_STRONGER_DECODER_GAP_SUPPORTED` but passes its pooled $\geq 4\times$ gate at exactly the boundary (11 of 20 draws) and carries no claim authority; P11F is diagnostic after a protocol-conformance defect; P11G is separately frozen with replay as a hard terminal gate.

Predictive `V`-information is the closest information-theoretic parent. Partial evaluation and knowledge compilation are the closest upstream-computation parents. Materialized views and multi-query optimization are reuse/crossover analogues. Modern query-conditioned memory, retrieval and context compression demonstrate practical task-conditioned state construction. Wong et al. provide current evidence that state design changes dynamic LLM reasoning. Sparse estimators and nonlinear ensembles are natural downstream substitutes.

P11's residual is not any primitive in isolation. It is the experimentally attacked relation between **where structure is exposed, how much discovery work remains downstream, what that exposure costs, and what future options it destroys or preserves**.

## 8 Limitations, discussion and conclusion

The exact theorem is restricted to a declared linear-access class and is not an unrestricted representation or time lower bound. Parity families are deliberately controlled. P11D shows that stronger decoder bias materially reduces compilation advantage. P11G tests one finite nonlinear ensemble. The current compiler is oracle/query-structured rather than learned. Controlled operation counts do not replace end-to-end latency, memory traffic, training cost or energy. Future-query laws assume the declared workload model. Broad real-system superiority still requires matched same-information experiments in real LLM/procedural and formal/search or long-horizon settings.

The experiments change the interpretation of task-conditioned construction. The strongest story is not that a compiler creates information or universally beats a universal representation. The decisive variable is **who performs structural search**. A query-conditioned compiler performs some search before the downstream learner sees state. A sparse decoder can recover part of that work; a deterministic finite tree ensemble is another search mechanism but remains inefficient under the registered high-dimensional low-sample envelope.

Aggressive specialization can make one task cheap while destroying immediate support for future tasks. Retaining raw state, caching compilations or materializing wider state are different allocations of computation and memory across time.

P11 establishes a controlled theory/systems result: fixed linear-accessible state scales with query-family rank; task-conditioned construction can expose smaller task-facing state; dense-decoder experiments show large gains; a hostile sparse decoder buys part of the gain back but leaves a reproducible $2\times/4\times$ residual; and a replay-gated deterministic nonlinear tree ensemble does not close the high-dimensional gap under its registered budget.

**State construction, decoder search, and future recoverability are coupled resource choices. Moving structure into state can reduce downstream dis-**

**covery work, but the benefit shrinks as the downstream access mechanism becomes better at discovering that structure, and specialization creates option debt unless recoverability is retained.**

The controlled claim does not authorize a universal nonlinear lower bound, transformer scaling law, free preprocessing or real-agent superiority.

# 9 References


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